Assignment 1

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July 19, 2025

To compare the performance of 10 convolutional neural networks (CNNs) pre-trained on the ImageNet dataset when applied to a subset of 20 classes from the CIFAR-100 dataset.

Dataset

- **Source:** CIFAR-100
- **Subset Used:** First 20 classes
- **Training Samples:** 4000
- **Testing Samples:** 1000
- **Image Size:** Originally 32x32, resized to 64x64 or 75x75 depending on model requirements

Models Evaluated

1. MobileNetV2
2. EfficientNetB0
3. VGG16
4. VGG19
5. ResNet50
6. ResNet101
7. InceptionV3
8. InceptionResNetV2
9. DenseNet121
10. Xception

Implementation Details

- **Transfer Learning:** Pre-trained base model (ImageNet) with frozen layers
- **Classification Head:** GlobalAveragePooling2D → Dense(64, relu) → Dropout(0.5) → Dense(20, softmax)
- **Optimizer:** Adam (0.001)
- **Loss Function:** Categorical Cross-Entropy
- **Epochs:** 3 (to conserve time/memory)
- **Batch Size:** 16
- **Evaluation Metric:** Accuracy on test set

Results Summary

Model	Accuracy	Loss	Time (s)	Parameters
Xception	**0.652**	1.1367	26.35	20,993,916
InceptionResNetV2	0.598	1.4212	119.56	54,436,404
MobileNetV2	0.568	1.4755	22.22	**2,341,268**
DenseNet121	0.563	1.3927	63.28	7,104,404
InceptionV3	0.515	1.6222	35.62	21,935,220
VGG16	0.487	1.7975	15.88	14,748,820
VGG19	0.407	2.0112	21.01	20,058,516
ResNet50	0.063	2.9859	37.94	23,720,148
ResNet101	0.052	2.9934	74.03	42,790,612
EfficientNetB0	0.045	2.9974	36.47	4,132,855

Analysis

- **Best Performance:** Xception achieved the highest accuracy (65.2%) with a moderate parameter size and efficient training time.
- **Efficient Choice:** MobileNetV2 provided a good trade-off between accuracy (56.8%), training time (22s), and very low model size (2.3M parameters).
- **Poor Performers:** ResNet50, ResNet101, and EfficientNetB0 performed poorly, possibly due to insufficient training epochs or ineffective upscaling from 32x32 images.
- **VGG Models:** Showed moderate accuracy but had high parameter counts, leading to inefficiency.

The complete source code used in this experiment is available on GitHub at the following link: Github